

Rigorous refraction law at the interface with an absorbing medium

Thibault J.-Y. Derrien, Jörg Krüger, Andreas Hertzwig, and Jörn Bonse
BAM Bundesanstalt für Materialforschung und -prüfung, Unter den Eichen 87, D-12205 Berlin, Germany
(Dated: April, 20th of 2014)

Snell-Descartes refraction law has still not a clear description in absorbing materials. Today, understanding transient optics involving intense laser irradiation and excitation of Surface Plasmon Polaritons under intense fields require a proper description of the refraction angle in absorbing materials, also while the approximation of low-absorbing materials drops down. In this work, several available models are compared with FDTD 3D simulations in absorbing materials and with ellipsometric phase measurements in reflection configuration.

CONTENTS

I. Introduction	1
A. The problem	1
B. The meaning of a complex angle	1
C. How to solve the problem ?	2
D. Could an experiment measure that ?	2
1. About measuring the angle	2
2. About measuring the reflected beam phase	2
E. So, is it so bad, doctor ?	2
F. Theoretical limit of Snell-Descartes law: mathematical and physical reasons	2
G. What if I just take the real part of the angle ?	3
H. More models	3
II. Numerical experiments	3
A. FDTD study of refracted angle as a function of incident angle in absorbing material	3
III. Experimental measurements	4
A. Phase reflectivity theory	4
B. Proposed experiment	4
References	4

I. INTRODUCTION

Despite that the problem of refraction law in absorbing materials is treated in many references from literature [1–4], no satisfying description was available in the case of strongly absorbing materials so far. Moreover, it is common to find references in which this difficulty is not mentioned despite that a proper description of refraction in strongly absorbing materials was required in both numerical [5] and experimental studies [6, 7].

The well-known law for refraction is given by Snell-Descartes law

$$\theta_{i+1} = \arcsin \left(\sqrt{\frac{\varepsilon_i}{\varepsilon_{i+1}}} \sin \theta_i \right).$$

Snell-Descartes law must be valid in case of complex dielectric functions as stated by Stratton [8] (and experimentally proved by ???). In case of total reflection,

a phase shift between equiphase plans and constant amplitude plans appears, leading to heterogeneous waves, then to interferences.

This work aims to correctly describe the refraction angle as a function of the incident angle and validate a general multilayer model, valid in both transparent and absorbing media.

A. The problem

At the interface with a slab material, a monochromatic wave experiments a reflection and a refraction as well described by the well-known Snell-Descartes law as follows for air environment

$$\sin \theta_i = n \sin \theta_r,$$

where $\theta_i \in \mathbb{R}$ is the incident angle, and θ_r is the refracted angle.

However, as soon as we consider an absorbing material, then, the optical index (and thus the dielectric permittivity $\varepsilon = n^2$) is a complex-valued quantity

$$n \in \mathbb{C}$$

and the angles of refraction also become a complex-valued quantity. As a consequence, the Snell law can remain valid, but only under certain conditions and an extended physical picture has to be considered.

B. The meaning of a complex angle

First, a complex angle has not a clear physical meaning. In the outer region of the sample, the incidence angle is well defined and real-valued. As soon as the wave gets into the material, the angles (both refraction and reflection) become complex-valued. Two interpretations can be given to a complex angle.

(i) Born and Wolf [1] considered it as a change of optical index as the angle is changing, which can be understood as anisotropy. However, nothing tells that simply using complex angles inside the sample is a sufficient formalism to properly consider this anisotropy.

(ii) A physical picture was proposed describing a shift between the equal amplitude wave front and the equal phase front.

In both cases, the angle of refraction cannot be considered as a well-defined peak value anymore and becomes a large distribution. A change of angle of refraction is nothing else than a change of the wave momentum, which is actually a *scattering event*. The laser wave is scattered by matter and interacts (at optical frequencies) with the electronic band structure. The case of an absorbing material requires to consider a rigorous approach to describe properly the wave scattering by an absorbing sample, for example by using transmission matrix formulation or direct solving of Maxwell equations. Anisotropic solution are found and a vector field formulation might be unavoidable. Recently, time-dependent Maxwell calculations have been performed in order to correctly reproduce the energy-dependent reflectivity measurements [9], and have identified a clear anisotropic response from the sample [10], giving explanation to arising anisotropic structures such as nanogratings in silica.

C. How to solve the problem ?

Before considering any advanced theories, it is proposed to consider the Snell Law using a multi-layered reflection model to calculate the sample reflectivity of a stratified material.

By comparison with a FDTD Maxwell 3D calculation, it will be shown that Snell-Descartes law can remain valid for low absorbing materials, but fails in the case of strongly absorbing materials. A simple criterion identifying the validity limit of the complex-valued Snell-Descartes law will be also proposed.

D. Could an experiment measure that ?

1. About measuring the angle

Not a simple one. Actually, the refracted angle can be measured if you can measure inside matter. Due to the backward light principle, the angle that is measured by reflection or by transmission through a thin-film is the same as the incident one, whatever happens inside the sample!

The way to produce such a measurement is thus to make the problem time-dependent and to observe a change of the refraction angle as a function of time ! This can be performed by an intense and ultrashort laser pulse which changes the optical properties of the sample as a function of time, and observe the transmitted wave direction evolution (how ?).

2. About measuring the reflected beam phase

However, by considering a multilayer model to calculate the surface reflectivity, it would be possible to obtain the phase of the reflected signal theoretically. The

formula involves the transmission angles inside the material, and complex optical indices. As a consequence, it is possible to compare the phase of the reflected beam with the phase of the calculated one. A comparison with FDTD calculation is also possible, despite that I focused on obtained the angle instead of the phase of the reflected beam.

E. So, is it so bad, doctor ?

So it seems that this theoretical limit of the Snell law is a real physical problem, and that the concept of refractive angle inside matter is not well understood yet for any materials. However, in the theoretical world (so in simulation codes!), this angle is required, and is still a complex-valued quantity. So this problem is important and must be solved in a clear way.

F. Theoretical limit of Snell-Descartes law: mathematical and physical reasons

Let's investigate now why Snell-Descartes law is limited to low-absorbing materials.

The Snell law is expressed by the following formula, using the dielectric permittivity

$$\sqrt{\varepsilon_i} \sin \theta_i = \sqrt{\varepsilon_{i+1}} \sin \theta_{i+1}$$

where i is the i -th layer of the sample. The complex-valued refractive angle θ_{i+1} is given by

$$\theta_{i+1} = \sin^{-1} \left[\sin \theta_i \sqrt{\frac{\varepsilon_i}{\varepsilon_{i+1}}} \right]$$

The real-valued arcsin function applied to $x \in \mathbb{R}$ has the following property

$$|\arcsin(x)| \leq \frac{\pi}{2}$$

However, the complex-valued number $\arcsin(z)$, where $z \in \mathbb{C}$ has a modulus which loses its trigonometric properties as soon as

$$\left| \sqrt{\frac{\varepsilon_i}{\varepsilon_{i+1}}} \right| > 1$$

This situation actually leads to a loss of the angle definition and constitutes a criterion which allows to check if the Snell-Descartes law can give an accurate description in an absorbing medium. This is the mathematical reason.

The physical reason is that the angle of refraction does not stand any longer since scattering occurs and that the angle becomes a broad distribution, which should be taken into account to conserve the energy and momentum.

G. What if I just take the real part of the angle ?

Let's do it. If we consider to take the real part of the ratio

$$\Re \sqrt{\frac{\varepsilon_i}{\varepsilon_{i+1}}},$$

then, indeed, the modulus of the sinus content is limited to the unity, and then no mathematical problem emerges from the model. However, this description is clearly not physical and is not valid in absorbing materials as demonstrated in the next section.

H. More models

Other references are available here :

M. H. Arzelies, La Réflexion Vitreuse, Annales de Physique (Paris), Vol. 1, 5-69 (1946)

I. N. Court, F. K. von Willisen, "Frustrated Total Internal Reflection and Application of its Principles to Laser Cavity Design", Applied Optics, Vol. 3, No. 6, 719-726 (1964)

P. E. Cidor, Refraction into an absorbing medium, American Journal of Physics 44, 786 (1976)

T. Oates, Progress on Surface Science 86 (2011)

II. NUMERICAL EXPERIMENTS

A. FDTD study of refracted angle as a function of incident angle in absorbing material

FDTD simulations have been performed in order to clarify the real refraction angle as a function of the material dielectric functions. The optical index $n + ik$ have been swept. Figure 1 presents the refracted angle as a function of real optical index n , and for a non-absorbing material ($k = 0$).

Figure 1 demonstrates the Snell-Descartes law in the case of a non-absorbing medium. All the theories fit to this case, but Euler equation fails at low optical index.

Figure 2 demonstrates the refraction angle as a function of the real part of the refractive index n for a imaginary part $k = 0.5$. It is shown that no model fit to the FDTD simulations data. Actually, the proposed results are already in the range that approximation $\Im m(\tilde{n}) \ll \Re e(\tilde{n})$ fails.

This discrepancy is also demonstrated for strongly absorbing materials ($k = 1$) on Figure 3. Snell law is not valid in the case of absorbing material, thus a special treatment has to be performed, as stated by Stratton [8].

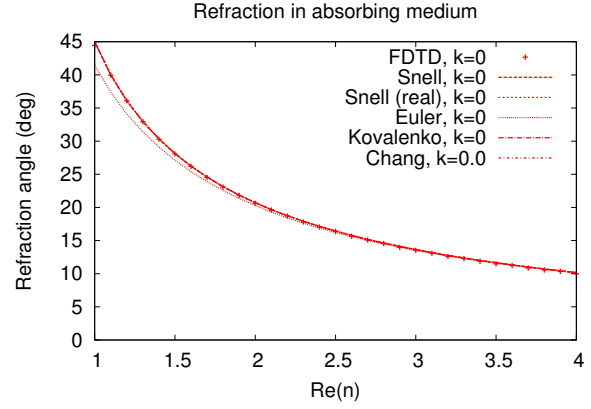


Figure 1. Comparison between FDTD simulation of the refraction angle with Snell law calculated in the case of a non-absorbing medium ($k = 0$). Incidence angle is 45° .

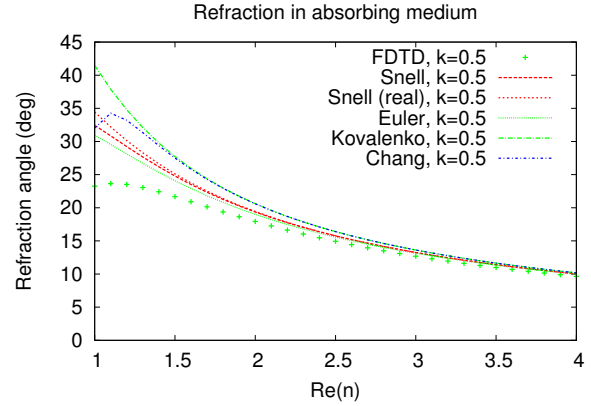


Figure 2. Comparison between FDTD simulation of the refraction angle with Snell law calculated in the case of an absorbing medium ($k = 0.5$). Incidence angle is 45° .

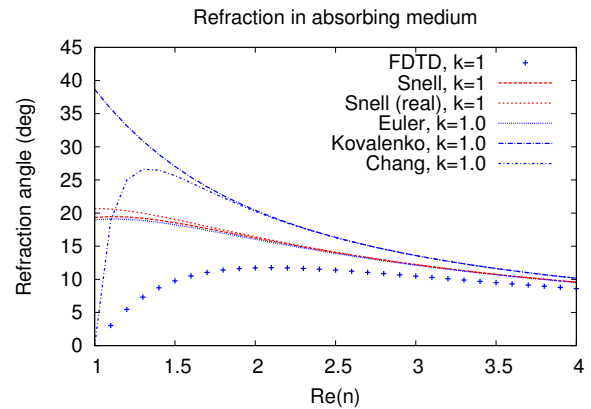


Figure 3. Comparison between FDTD simulation of the refraction angle with Snell law calculated in the case of a strongly absorbing medium ($k = 1.0$).

III. EXPERIMENTAL MEASUREMENTS

A. Phase reflectivity theory

Surface reflectivity is defined by $R = |r_{0,N}|^2$ where N is the number of considered optical layers and layer 0 corresponds to the external environment. The surface reflectivity is described by the the Fresnel equations [1, 11]. We take into account the contribution of each layer to the surface reflectivity by using a recurrence formula given by the transmission matrix model [1, 5, 12]

$$r_{i,k} = \frac{r_{i,i+1} + r_{i+1,k} e^{2i\phi_{i+1}}}{1 + r_{i,i+1} r_{i+1,k} e^{2i\phi_{i+1}}} \quad (1)$$

where

$$\phi_{i+1} = \frac{2\pi h_{i+1}}{\lambda} \sqrt{\varepsilon_{i+1}} \cos \theta_{i+1} \quad (2)$$

is the phase shift induced by the $i + 1$ layer, h_{i+1} is the thickness of the layer $i + 1$, and θ_{i+1} is the angle of inci-

dence at layer $i + 1$. $r_{i,i+1}$ is given by Fresnel formulas [13]

$$r_{i,i+1} = \frac{\sqrt{\varepsilon_i} \cos \theta_i - \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}}{\sqrt{\varepsilon_i} \cos \theta_i + \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}} \quad (3)$$

in the case of perpendicular polarization, and

$$r_{i,i+1} = \frac{\varepsilon_{i+1} \cos \theta_i - \sqrt{\varepsilon_i} \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}}{\varepsilon_{i+1} \cos \theta_i + \sqrt{\varepsilon_i} \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}} \quad (4)$$

in the case of a parallel polarization. ε_i is the full complex dielectric function calculated at the layer i , and θ_i is the angle of incidence at the layer i .

B. Proposed experiment

What can be verified experimentally is the knowledge of the refractive angle θ_{i+1} in Eq. (2). This can be performed by calculating the phase of a perfect slab sample.

-
- [1] M. Born and E. Wolf. *Principles of Optics. Electromagnetic theory of propagation, interference and diffraction of light.* (Cambridge University Press, 1980), 7th edition edition.
 - [2] E. D. Palik. *Handbook of Optical Constants of Solids* (Academic Press, 1985).
 - [3] S. Kovalenko. *Semiconductor Physics* **4**, p. 3 (2001).
 - [4] J. Shen, H. Yu, and J. Lu. *Chinese Optics Letters* **8**, p. 1 (2010).
 - [5] N. M. Bulgakova, R. Stoian, and A. Rosenfeld. *Quantum Electron.* **40**, 11, pp. 966–985 (2010).
 - [6] K. Sokolowski-Tinten and D. von der Linde. *Phys. Rev. B* **61**, pp. 2643–2650 (2000).
 - [7] C. Shank, R. Yen, and C. Hirlimann. *Phys. Rev. Lett.* **50**, 6, pp. 454 – 457 (1983).
 - [8] J. A. Stratton. *Electromagnetic Theory* (McGraw-Hill Book Company, 1941).
 - [9] H. Zhang, S. A. Wolbers, D. M. Krol, J. I. Dijkhuis, and D. van Oosten. *arXiv preprint arXiv:1402.4998* (2014).
 - [10] N. M. Bulgakova, V. P. Zhukov, and Y. P. Meshcheryakov. *Applied Physics B* **Published online** (2013).
 - [11] A. Prokhorov, V. Konov, I. Ursu, and I. Mihailescu. *Laser Heating of Metals* (Adam Hilger, 1990).
 - [12] D. Bäuerle. *Laser Processing and Chemistry* (Springer-Verlag, 2011), 4th edition edition.
 - [13] J. D. Jackson. *Classical Electrodynamics* (Wiley, 1999), 3rd edition edition.